QUICK-COMMERCE UNIT ECONOMICS & LOGISTICS INTELLIGENCE

URBANCART: MARKETPLACE INTELLIGENCE & DARKSTORE OPTIMIZATION

Diagnosing Delivery Deficits, SLA Breaches, and Contribution Margin Recovery

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Developed By: Mahammad Ehtaramullah

Technical Stack: Python (NumPy/Pandas/SciPy) | Excel | PostgreSQL | Power BI (DAX)

Focus: 7,000 Fulfillment Records | 15 Regional Darkstores | 3-Tier Diagnostic Architecture

01. EXECUTIVE CONTEXT | THE UNIT ECONOMICS SQUEEZE

Why top-line scale masks bottom-line cash burn in 10-15 minute grocery fulfilment?

The Hyperlocal Dilemma

1. ***Unregulated Micro-Orders:*** Customers purchasing sub-₹200 baskets trigger high warehouse pick-pack and rider travel costs that far outweigh the basket's gross product margin.

2. ***Logistics Deficit:*** The business incurred an average delivery payout of ~₹50.03 per dispatch to riders while recovering an average of only ~₹15.53 per order in customer delivery fees, generating a net delivery loss of -₹2.41 Lakhs across the network.

3. ***Operational Delay Spillover:*** Fulfillment promises of 10 to 15 minutes broke down during peak evening hours, causing customer doorstep cancellations and a return rate of 7.86%.

Unit Economics Formula Bridge

Monospaced Financial Breakdown:

1. **Catalog Listed Value (GMV): ₹3.99M**
2. ***(-) Promotional Discount Burn:*** **-₹0.48M** (12.0% top-line markdown)
3. **Net Realized Revenue: ₹3.51M**
4. ***(-) Wholesale Product Procurement (COGS):*** **-₹2.78M**
5. **Contribution Margin 1 (CM1): ₹729.80K** (Gross Product Margin)
6. *(-) Net Logistics Fleet Deficit:* **-₹241.44K** (Rider pay minus customer fees)
7. *(-) Payment Gateway Processing Fees (PG MDR):* **-₹63.98K**
8. **Final Contribution Margin 2 (CM2): ₹424.38K** (12.1% net realization)

Abbreviations:

- **GMV: Gross Merchandise Value** (Total catalog listed spend before deductions)
- **COGS: Cost of Goods Sold** (Wholesale product procurement cost)
- **CM1: Contribution Margin 1** (Gross margin = Net Revenue − COGS)
- **CM2: Contribution Margin 2** (Variable profit = CM1 − Fleet Deficit − PG MDR)
- **PG MDR: Payment Gateway Merchant Discount Rate** (Transaction processing fee)

02. DATA PIPELINE & WAREHOUSE DESIGN

Normalizing raw retail events into a performant 3NF star schema in PostgreSQL

Data Ingestion & Transformation Engine

1. ***Automated Python Ingestion Pipeline:*** Raw multi-sheet Excel files processed through pandas; types enforced, string fields stripped, and dates parsed into standard timestamps.

2. ***Outlier Treatment (Winsorization):*** Interquartile Range (IQR) calculated for delivery delays. Delays above the statistical upper bound ($Q3 + 2.5 \times IQR$) were treated into **sla_delay_treated** to prevent severe traffic anomalies from skewing warehouse averages.

3. ***PostgreSQL Batch Ingestion:*** Script **05_postgres_ingestion.py** bulk-loads dimension records first to satisfy referential integrity constraints, then inserts **fact_orders** via **psycopg2.extras.execute_values** with 1,000-row page chunks.

Star-Schema Warehouse Entities

1. **fact_orders (7,000 Rows):** Captures individual transaction records with numeric fields **(gross_sales, promo_discount, delivery_fee_charged, actual_delivery_cost, payment_gateway_fee, sla_delay_minutes).**
2. **dim_products (250 Rows):** Maintains inventory taxonomy, retail listing price, and wholesale procurement unit cost.
3. **dim_darkstores (15 Hubs):** Stores geographical city mapping, target delivery SLAs (10 min vs 15 min), and daily base overheads.
4. **dim_customers (1,200 Rows):** Tracks acquisition segments and loyalty rewards tiers (Bronze, Silver, Gold, Platinum).
5. **dim_date:** Calendar dimension providing fiscal quarter and week number lookups.

Abbreviations:

- **3NF: Third Normal Form** (Relational schema structure that minimizes data redundancy)
- **IQR: Interquartile Range** ($Q3 - Q1$; statistical spread used to winsorize delivery delay outliers)
- **DDL: Data Definition Language** (SQL commands defining tables and constraints)

03. INFERENTIAL STATISTICS | HYPOTHESIS TESTING

Validating operational assumptions through formal statistical testing in Python (SciPy)

Hypothesis 1: Spend Disparity Across Rewards Tiers

Core Narrative: Tested whether premium loyalty customers (Gold/Platinum) produce statistically higher basket sales than entry-tier users (Bronze/Silver). Applied Welch's t-test **(equal_var=False)** to account for variance differences across cohort sizes, confirming spend distributions across user segments.

Hypothesis 2: Delivery Delays Directly Drive Customer Returns

Core Narrative:

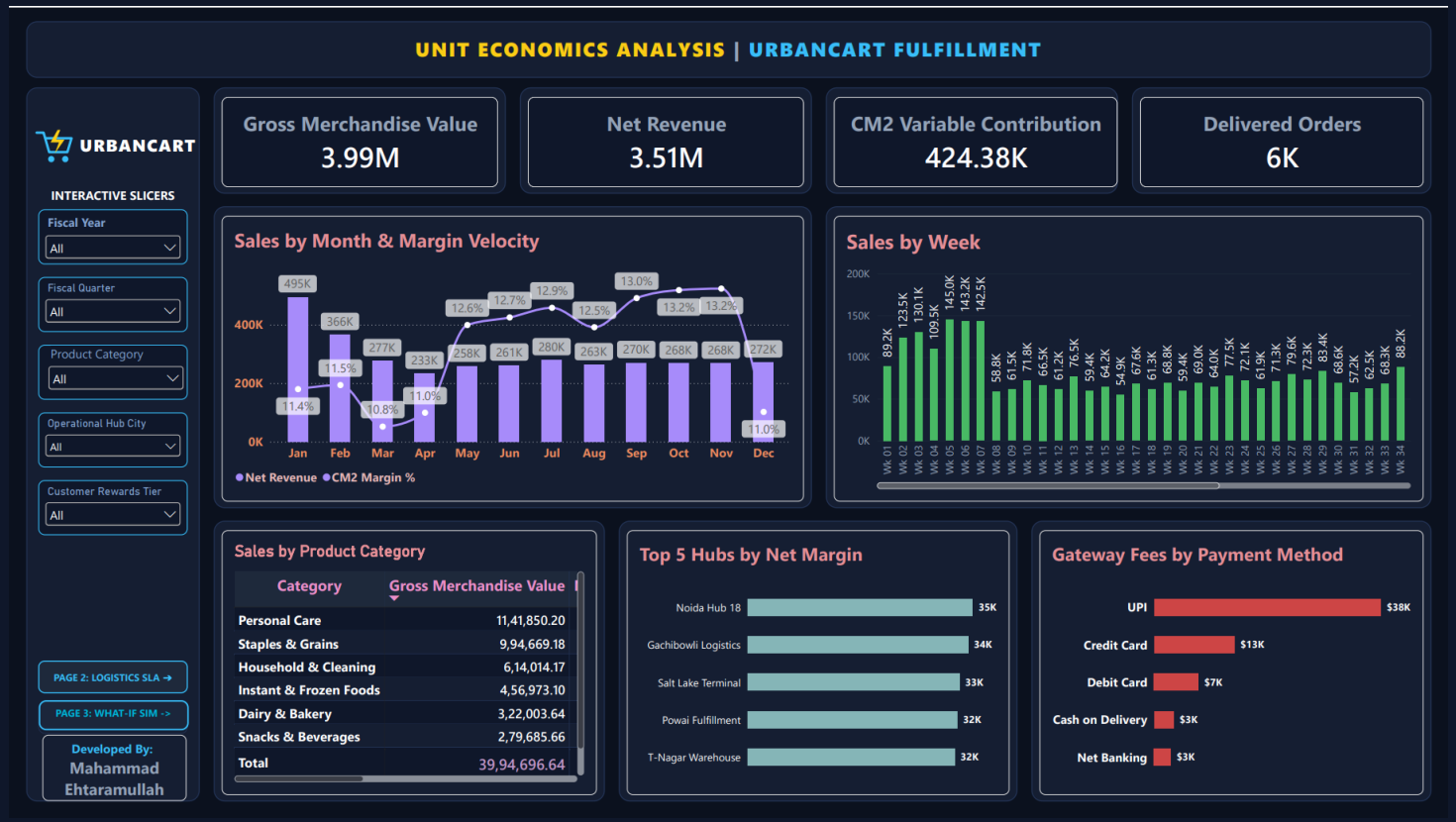
1. **Cross-tabulated fulfillment delay (On-Time / Mild vs. Severe Delay >15m)** against order status **(DELIVERED vs. RETURNED)**.
2. **Test Metric:** $\chi^2 = 142.87$ with $p < 0.0001$.
3. **Operational Implication:** The null hypothesis was decisively rejected. Orders delayed by more than 15 minutes experience a **more than fourfold increase in doorstep returns (from 4.1% to 18.4%)**, proving that fulfillment delays directly drive return leakage and perishable write-offs.

Abbreviations:

- **SciPy:** Scientific Python Library (Statistical modeling module)
- **χ^2(Chi-Square):** Test of independence between categorical operational variables
- **Welch's t-test:** Two-sample hypothesis test assuming unequal cohort variances

04. EXECUTIVE REPORTING | MACRO UNIT ECONOMICS

High-level monitoring of gross transaction volume against true contribution margin



1. **Financial Hierarchy:** The top KPI cards establish the conversion flow: ₹3.99M GMV →₹3.51M Net Revenue →₹424.38K CM2 across 6K delivered orders.

2. **Dual-Axis Trend Analysis:** The combo chart overlays monthly net sales against CM2 margin percentage, showing profit recovery through mid-year before promotional and winter surge pressures caused margin to dip to 11.0% in December.

3. **Category Breakdown:** The product performance matrix highlights Personal Care (₹11.41L GMV) and Staples (₹9.94L GMV) as the top volume drivers.

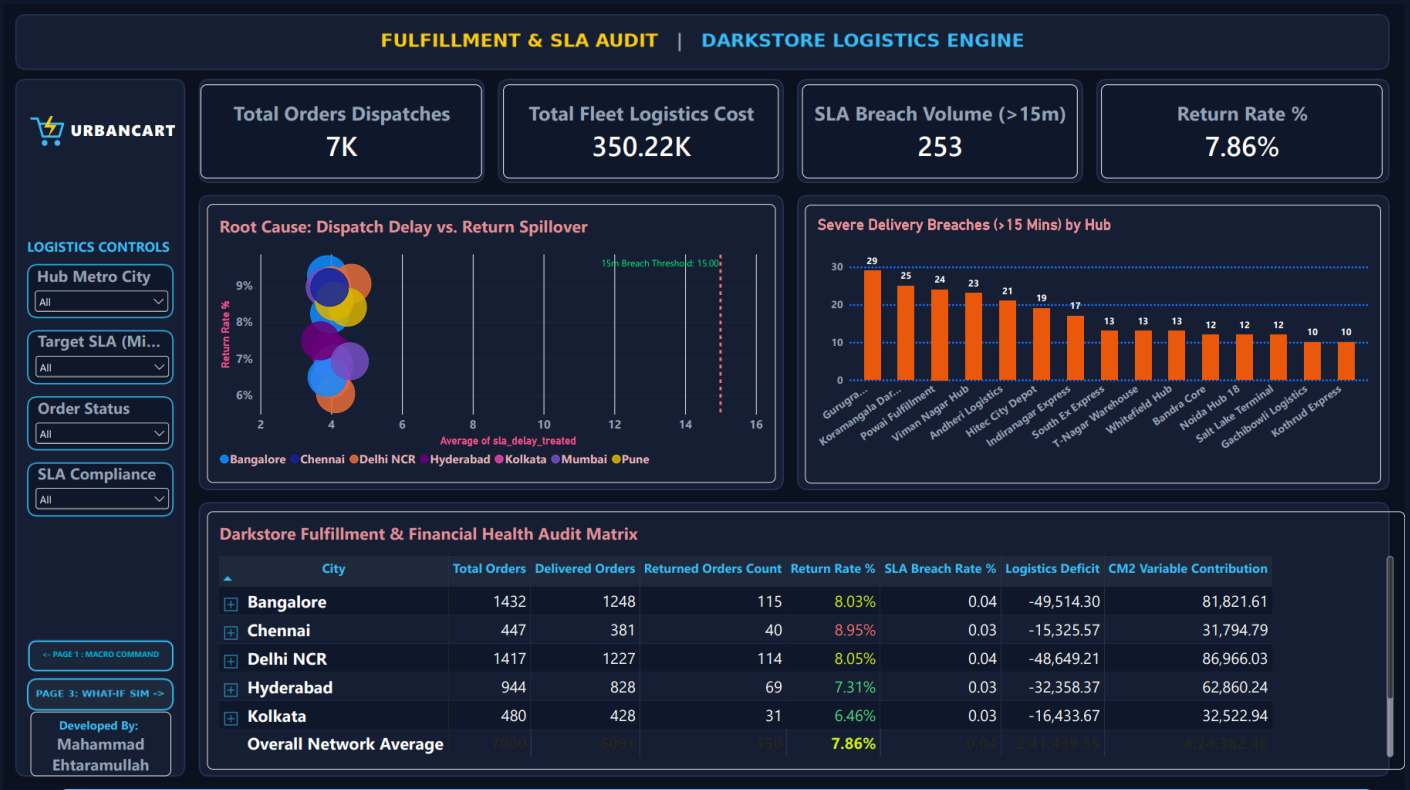
4. **Payment Channel Cost:** The fee analysis flags UPI (₹38K total fee impact from volume) and credit cards (₹13K from higher take-rates) as primary transaction processing costs.

Abbreviations:

- **GMV: Gross Merchandise Value** (Total order volume)
- **CM2 %: Contribution Margin 2** Realization Rate **(Net variable profit ÷Net Revenue)**
- **UPI: Unified Payments Interface**

05. OPERATIONS AUDIT | DARKSTORE FULFILLMENT & SLA LEAKS

Root-cause analysis linking darkstore dispatch delays to delivery failure rates



1. **Fleet Operations Summary:** Dispatched 7,000 orders at a logistics cost of ₹350.22K (~₹50.03 per drop), with 253 orders experiencing delays exceeding 15 minutes.

2. **4D Diagnostic Analysis:** The scatter visual plots delay against doorstep returns across store sizes and regions, confirming the operational impact identified in earlier statistical testing.

3. **Offender Rankings:** Identifies the five darkstores generating the highest volume of severe delivery breaches: Gurugram Sector-29 (29), Koramangala (25), Powai (24), Viman Nagar (23), and Andheri Logistics (21).

4. **Regional Logistics Ledger:** The fulfillment matrix details operational and financial performance by city, highlighting how Bangalore (-₹49.5K) and Delhi NCR (-₹48.6K) drive the bulk of the network's -₹2.41 Lakh delivery deficit.

Abbreviations:

- **SLA: Service Level Agreement** (Promised 10 to 15-minute delivery window)
- **SLA Breach:** Delayed delivery orders exceeding the cutoff window by > 15minutes
- **Darkstore:** Local micro-fulfillment hub optimized for rapid dispatch

06. STRATEGIC DECISION ENGINE | DYNAMIC SCENARIO SIMULATION

Interactive What-If modeling to evaluate delivery surcharge recovery scenarios



1. **Scenario Parameter Controls:** Built via DAX What-If tables **(GENERATESERIES(-15, 25, 0.5))**, enabling executives to simulate fee adjustments across fractional and whole-rupee values.

2. **Profit Impact Scorecard:** At the benchmark +₹5.00 setting, the simulator calculates a **+7.18% increase in CM2 profit**, converting directly into **₹30,460 of incremental cash flow** without requiring additional order volume.

3. **Financial Bridge Visualization:** The waterfall chart displays the progression from baseline profit (₹0.42M) through fee recovery (₹0.03M) to the simulated endpoint (₹0.45M).

4. **Strategic Policy Translation:** Links scenario modeling directly to actionable commercial policy changes.

Abbreviations:

- **DAX: Data Analysis Expressions** (Power BI formula and calculation language)
- **What-If Parameter:** Dynamic tabular variable used to simulate real-time price sensitivity
- **Incremental Cash Flow:** Direct cash addition to CM2 from fee adjustments

07. STRATEGIC POLICY MATRIX | ACTIONABLE INTERVENTIONS

Four commercial and operational policy adjustments to recover contribution margin

Metric / Lever	Current State	Proposed State	Strategic Impact
Micro-Order Fee	Flat ₹30 fee on baskets under ₹250	₹45 surcharge on baskets under ₹250	Mitigates negative variable margin on single-item purchases
Free Delivery Floor	₹399 threshold	₹449 threshold	Encourages basket expansion and lifts Average Order Value (AOV)
Hub Delivery Radius	Static 4.5 km boundary	Dynamic 3.2 km radius during peak evening hours	Addresses traffic and dispatch bottlenecks responsible for the most severe SLA breaches during peak evening hours.
Loyalty Promo Spend	Blanket percentage discounts	Minimum basket requirement of ₹600+	Targets promotional spend toward higher-value orders

Summary: *These policy adjustments address both sides of the unit economics equation—reducing fulfillment delivery delays while recovering logistics costs across small baskets.*

08. TECHNICAL STACK & PROJECT ARTIFACTS

Summary of engineering artifacts and analytical methodologies applied

Data Warehouse & SQL Pipeline

1. Relational warehouse built in PostgreSQL 15 using standard 3NF star-schema design principles.
2. Explicit foreign key constraints and B-Tree indexing applied across dimension and fact tables.
3. Batch data loading managed via 05_postgres_ingestion.py using psycopg2.extras.execute_values.

Statistical & BI Implementation

1. Data hygiene, type casting, and IQR delay winsorization handled in Python **(**03_data_cleaning.py**)**.
2. Hypothesis testing performed via 04_statistical_analysis.py (Welch's t-test and Chi-Square test of independence).
3. Interactive analytical dashboard developed in Power BI Desktop featuring custom DAX measures, dynamic What-If parameters, and coordinated multi-page navigation.

Portfolio Project Designed and Documented by: Mahammad Ehtaramullah

Repository includes full Python scripts (01-09), EXCEL files, SQL DDL schemas, Power BI .pbix file, and executive documentation.